

Numerical Computing-II — Assignment-II

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1. Objective

This assignment investigates the accuracy of three finite-difference formulas — Forward, Backward, and Central difference — for approximating the first derivative. We compare their performance on three test functions across six step sizes h , evaluate $f'(1)$ for each, compute the absolute error against the exact analytical derivative, and identify which method is most reliable and at what point round-off error dominates.

2. Algorithm / Pseudocode

Before implementation, the overall procedure and each finite-difference rule are stated as pseudocode.

2.1 Overall Algorithm

ALGORITHM CompareDifferentiationMethods

INPUT : point $x_0 = 1$

set of test functions $F = \{ e^x, \sin(x), x^3 - 2x + 1 \}$

set of step sizes $H = \{ 1e-1, 1e-2, 1e-3, 1e-4, 1e-5, 1e-6 \}$

OUTPUT: results table, log-log error plot

FOR each function f in F :

exact $\leftarrow$ AnalyticalDerivative(f, x_0)

FOR each h in H :

$F_h \leftarrow$ ForwardDifference(f, x_0, h)

$B_h \leftarrow$ BackwardDifference(f, x_0, h)

$C_h \leftarrow$ CentralDifference(f, x_0, h)

$errF \leftarrow | \text{exact} - F_h |$

$errB \leftarrow | \text{exact} - B_h |$

$errC \leftarrow | \text{exact} - C_h |$

RECORD ($f, h, F_h, B_h, C_h, errF, errB, errC$)

END FOR

END FOR

PLOT $\log_{10}(h)$ vs $\log_{10}(errF)$, $\log_{10}(errB)$, $\log_{10}(errC)$ for every f

INSPECT slopes $\rightarrow$ confirm $O(h)$ for Forward/Backward, $O(h^2)$ for Central

IDENTIFY h below which error stops decreasing (round-off floor)

2.2 Forward Difference

FUNCTION ForwardDifference(f, x, h)

RETURN ($f(x + h) - f(x)$) / h

END FUNCTION

Uses $f(x)$ and one point ahead of x . Truncation error is $O(h)$.

2.3 Backward Difference

FUNCTION BackwardDifference(f, x, h)

RETURN ($f(x) - f(x - h)$) / h

END FUNCTION

Uses $f(x)$ and one point behind x . Truncation error is $O(h)$.

2.4 Central Difference

FUNCTION CentralDifference(f, x, h)

RETURN (f(x + h) - f(x - h)) / (2 * h)

END FUNCTION

Uses one point ahead and one point behind x. Truncation error is $O(h^2)$, i.e. it converges quadratically -> generally the most accurate for a given h.

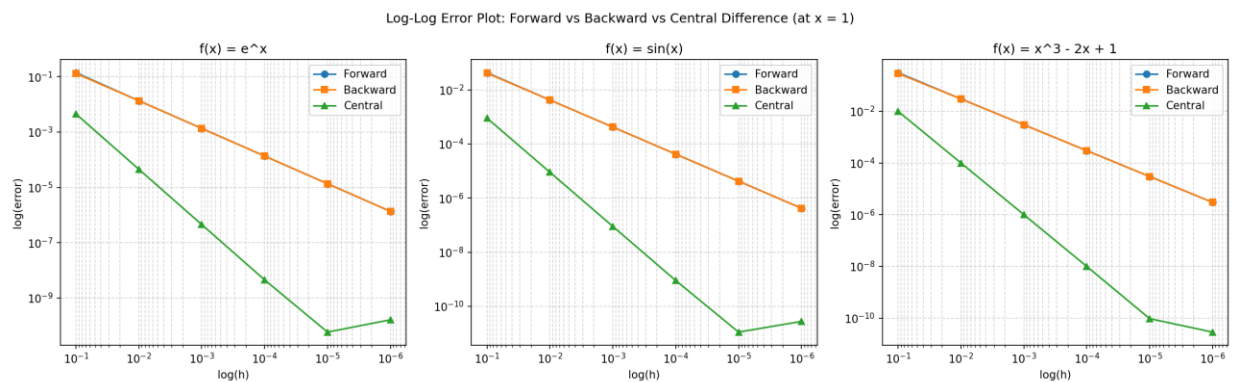
4. Results Table

$f'(1)$ approximated by each method, and the absolute error against the exact analytical derivative, for $h = 10^{-1} \dots 10^{-6}$:

Function	h	Forward f'	Backward f'	Central f'	Err Forward	Err Backward	Err Central
$f(x) = e^x$	1.000e-1	2.859e+0	2.587e+0	2.723e+0	1.406e-1	1.315e-1	4.533e-3
$f(x) = e^x$	1.000e-2	2.732e+0	2.705e+0	2.718e+0	1.364e-2	1.355e-2	4.530e-5
$f(x) = e^x$	1.000e-3	2.720e+0	2.717e+0	2.718e+0	1.360e-3	1.359e-3	4.530e-7
$f(x) = e^x$	1.000e-4	2.718e+0	2.718e+0	2.718e+0	1.359e-4	1.359e-4	4.531e-9
$f(x) = e^x$	1.000e-5	2.718e+0	2.718e+0	2.718e+0	1.359e-5	1.359e-5	5.859e-11
$f(x) = e^x$	1.000e-6	2.718e+0	2.718e+0	2.718e+0	1.359e-6	1.359e-6	1.635e-10
$f(x) = \sin(x)$	1.000e-1	4.974e-1	5.814e-1	5.394e-1	4.294e-2	4.114e-2	9.001e-4
$f(x) = \sin(x)$	1.000e-2	5.361e-1	5.445e-1	5.403e-1	4.216e-3	4.198e-3	9.005e-6
$f(x) = \sin(x)$	1.000e-3	5.399e-1	5.407e-1	5.403e-1	4.208e-4	4.206e-4	9.005e-8
$f(x) = \sin(x)$	1.000e-4	5.403e-1	5.403e-1	5.403e-1	4.207e-5	4.207e-5	9.004e-10
$f(x) = \sin(x)$	1.000e-5	5.403e-1	5.403e-1	5.403e-1	4.207e-6	4.207e-6	1.114e-11
$f(x) = \sin(x)$	1.000e-6	5.403e-1	5.403e-1	5.403e-1	4.207e-7	4.208e-7	2.772e-11
$f(x) = x^3 - 2x + 1$	1.000e-1	1.310e+0	7.100e-1	1.010e+0	3.100e-1	2.900e-1	1.000e-2
$f(x) = x^3 - 2x + 1$	1.000e-2	1.030e+0	9.701e-1	1.000e+0	3.010e-2	2.990e-2	1.000e-4
$f(x) = x^3 - 2x + 1$	1.000e-3	1.003e+0	9.970e-1	1.000e+0	3.001e-3	2.999e-3	1.000e-6
$f(x) = x^3 - 2x + 1$	1.000e-4	1.000e+0	9.997e-1	1.000e+0	3.000e-4	3.000e-4	1.000e-8
$f(x) = x^3 - 2x + 1$	1.000e-5	1.000e+0	1.000e+0	1.000e+0	3.000e-5	3.000e-5	9.537e-11
$f(x) = x^3 - 2x + 1$	1.000e-6	1.000e+0	1.000e+0	1.000e+0	3.000e-6	3.000e-6	2.876e-11

5. Log-Log Error Plot

Generated directly from the C++ program's CSV output using Python/Matplotlib (log-log axes):



6. Discussion

6.1 Which method is consistently the most accurate?

Central difference is consistently the most accurate across all three functions and every h tested (before round-off dominates). Its error is roughly two orders of magnitude smaller than Forward/Backward at the same h — e.g. for e^x at $h = 10^{-2}$, Central error $\approx 4.5 \times 10^{-5}$ versus $\approx 1.4 \times 10^{-2}$ for Forward/Backward. This matches theory: Central is $O(h^2)$ while Forward and Backward are only $O(h)$.

6.2 Does the observed error decrease at the expected rate?

Yes. On the log-log plot, Forward and Backward errors fall on lines of slope ≈ 1 (error $\propto h$), while Central error falls on a line of slope ≈ 2 (error $\propto h^2$) — visible as the steeper central-difference lines in Section 5. This is confirmed exactly for the polynomial in Section 3.4, where the error is a closed-form expression linear in h (Forward/Backward) or exactly quadratic in h (Central).

6.3 Is there a value of h beyond which the error stops decreasing?

Yes. Below roughly $h \approx 10^{-5}$ – 10^{-6} the Central-difference curves flatten and even tick back up slightly instead of continuing to fall — visible as the plateau at the bottom-left of the log-log plot. Truncation error keeps shrinking as $h \rightarrow 0$, but subtracting two nearly-equal floating point numbers $f(x+h)$ and $f(x-h)$ (or $f(x+h)$ and $f(x)$) loses significant digits to round-off (machine epsilon $\approx 2.2 \times 10^{-16}$ for double precision). Beyond that point round-off error dominates truncation error, so total error stops decreasing and can even increase again for smaller h . Central difference reaches this floor sooner because its truncation term is already tiny — a practical limit on how small h should be chosen. An 'optimal' h balances truncation error ($\propto h^2$) against round-off error ($\propto \epsilon/h$).